

```
#!/usr/bin/python
# -*- coding:utf-8 -*-
import sys
import os
libdir = os.path.join(os.path.dirname(os.path.
dirname(os.path.realpath(__file__))), 'lib')
if os.path.exists(libdir):
    sys.path.append(libdir)

from TP_lib import gt1151
from TP_lib import epd2in13_V2
import time
import logging
import traceback
import threading
from pynput.keyboard import Key, Controller

|
logging.basicConfig(level=logging.DEBUG)
flag_t = 1
keyboard = Controller()

def pthread_irq():
    print("pthread running")
    while flag_t == 1:
        if(gt.digital_read(gt.INT) == 0):
            GT_Dev.Touch = 1
        else:
            GT_Dev.Touch = 0
        print("thread:exit")

try:
    logging.info("epd2in13_V2 Touch Demo")
```

Fig. 13: Code for keyboard input

```
#uncomment to overclock the arm. 700 MHz is the default.
#arm_freq=800

# Uncomment some or all of these to enable the optional hardware interfaces
dtparam=i2c_arm=on
#dtparam=i2s=on
dtparam=spi=on
enable_uart=1
dtoverlay=spi1-3cs,cs0_pin=05,cs1_pin=06,cs2_pin=16

# Uncomment this to enable the lirc-rpi module
#dtoverlay=lirc-rpi

# Additional overlays and parameters are documented /boot/overlays/README

[ Enable audio (loads snd_bcm2835)
dtparam=audio=on
dtoverlay=waveshare35a

AG Get Help  AO Write Out  AW Where Is  AK Cut Text  AJ Justify  AC Cur Pos
AX Exit      AR Read File  AN Replace  AU Uncut Text AT To Spell  AL Go To Line
```

Fig. 16: Enabling other SPI on Raspberry Pi

```
File Edit Format Run Options Window Help

t = threading.Thread(target = pthread_irq)
t.setDaemon(True)
t.start()
while(1):

    gt.GT_Scan(GT_Dev, GT_Old)
    if(GT_Old.X[0] == GT_Dev.X[0] and GT_Old.Y[0] == GT_Dev.Y[0] and GT_Old.S[0] == GT_Dev.S[0]):
        continue
    if(GT_Dev.X[0] > 100 and GT_Dev.X[0] < 114 and GT_Dev.Y[0] > 2 and GT_Dev.Y[0] < 37):
        keyboard.type('1')
        print("1")
    if(GT_Dev.X[0] > 100 and GT_Dev.X[0] < 114 and GT_Dev.Y[0] > 38 and GT_Dev.Y[0] < 51):
        keyboard.type('2')
        print("2")

    elif(GT_Dev.X[0] > 100 and GT_Dev.X[0] < 114 and GT_Dev.Y[0] > 52 and GT_Dev.Y[0] < 71):
        keyboard.type('3')
        print("3")
    elif(GT_Dev.X[0] > 100 and GT_Dev.X[0] < 114 and GT_Dev.Y[0] > 73 and GT_Dev.Y[0] < 100):
        keyboard.type('4')
        print("4")
    elif(GT_Dev.X[0] > 100 and GT_Dev.X[0] < 114 and GT_Dev.Y[0] > 101 and GT_Dev.Y[0] < 123):
        keyboard.type('5')
        print("5")
    elif(GT_Dev.X[0] > 100 and GT_Dev.X[0] < 114 and GT_Dev.Y[0] > 125 and GT_Dev.Y[0] < 149):
        keyboard.type('6')
        print("6")
    elif(GT_Dev.X[0] > 100 and GT_Dev.X[0] < 114 and GT_Dev.Y[0] > 150 and GT_Dev.Y[0] < 172):
        keyboard.type('7')
        print("7")
    elif(GT_Dev.X[0] > 100 and GT_Dev.X[0] < 114 and GT_Dev.Y[0] > 175 and GT_Dev.Y[0] < 197):
        keyboard.type('8')
        print("8")
    elif(GT_Dev.X[0] > 100 and GT_Dev.X[0] < 114 and GT_Dev.Y[0] > 199 and GT_Dev.Y[0] < 223):
        keyboard.type('9')
        print("9")
    elif(GT_Dev.X[0] > 100 and GT_Dev.X[0] < 114 and GT_Dev.Y[0] > 225 and GT_Dev.Y[0] < 248):
```

Fig. 14: Code checking touchpoints and converting into keyboard input

```
#
import RPi.GPIO as GPIO
import time
from smbus import SMBus
import spidev
import ctypes
import logging

# e-Paper
EPD_RST_PIN = 17
EPD_DC_PIN = 25
EPD_CS_PIN = 8
EPD_BUSY_PIN = 24

# TP
TRST = 22
INT = 27

spi = spidev.SpiDev(1, 0)
address = 0x0
# address = 0x14
# address = 0x48
bus = SMBus(1)
```

Fig. 15: Changing the SPI number in code

TABLE 3
CONNECTION OF E-INK DISPLAY TO RASPBERRY PI

E-ink paper pins	BCM pin numbering	Board pin number
VCC	3.3V	3.3V
GND	GND	GND
INT	27	13
SCL	SCL1	5
SDA	SDA1	3
TRST	22	15

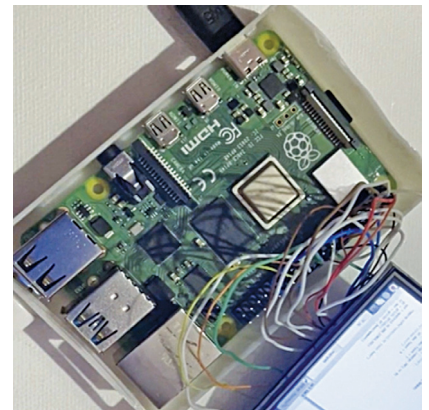


Fig. 17: Raspberry Pi fitted in the case

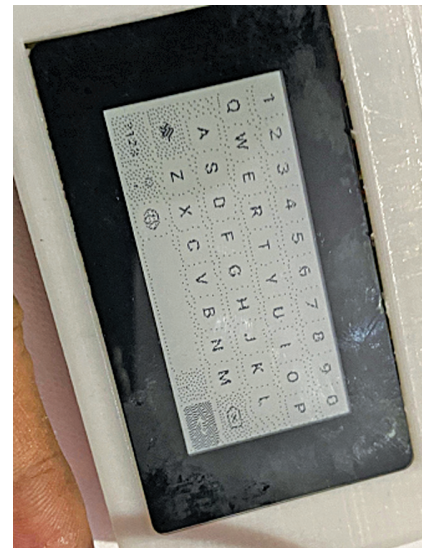


Fig. 18: Touch keyboard on the top of laptop base